036015

Finite Elements Method for Engineering Analysis

Homework Assignment 2 - 1D Linear Element

Winter 2026

Instructions

- Electronic submission of **typed** sheets only (PDF format through moodle).
- Submission file must be named: "*name surname.pdf*". No I.D number or homework number is required in the file name.
- Solution must contain all the relevant derivations, explanations, graphs and conclusions.
- All codes used must be added to the submission files.

Consider an elastic rod, fixed to a rotating hinge as shown in Figure 1:

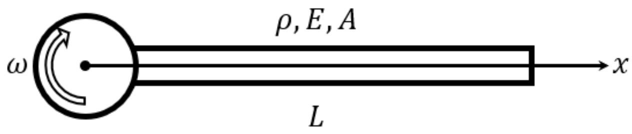

Figure 1: Uni-axially loaded rotating bar

where E is Young's modulus, A is the constant cross section area of the rod, L is the length of the rod, ρ is the constant density of the rod and ω is the angular velocity of the rotation.

Neglecting the bending of the rod, the BVP that governs the problem is given by:

$$\begin{cases} -\dfrac{d}{dx}\left(EA\dfrac{du}{dx}\right) = \rho A \omega^2 x \\ u(0) = 0 \\ EA\left.\dfrac{du}{dx}\right|_{x=L} = 0 \end{cases} \tag{1}$$

Where $u(x)$ is the stretch of the rod along the x axis during rotation as a result of the apparent outward force - the centrifugal force.

1. Solve the problem analytically (obtain the exact solution). [15pt]
2. Apply the least squares method to solve the problem at three points $(0, L/3, L)$, using Lagrange polynomials as basis functions. Report the three constant values obtained. Provide a detailed explanation of your approach, including all necessary derivations and calculations. [15pt]

The following part must be completed on paper (not coded):

3. Apply the Galerkin method and formulate the weak form of the problem. [15pt]
4. Based on the result from the previous section, write the integral form of the stiffness matrix of a master (local) element. [5pt]

5. Calculate the stiffness matrix of an element, using **local** linear Lagrange basis functions. Calculate the integral analytically. [15pt]

6. Interpolate the forcing function $f(x)$ using Lagrange polynomials and define and calculate the mass matrix of the master element. Write the forcing vector of an element $F_i = M_{ij} f_j$. [15pt]

7. Solve the problem using two elements as formulated in the previous sections. Use a uniform mesh (all elements of the same length). [20pt]

Good Luck!